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Going with the Flow: Variational Bayesian Inference using Flow Matching

by

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This is my own work (except where otherwise indicated)

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A handwritten signature in black ink, appearing to read 'Javůrek', written in a cursive style.

Date: 13 September 2024

Abstract

In this dissertation, we investigate Flow Matching for Posterior Estimation (FMPE) as the latest method for variational inference. FMPE learns a Continuous Normalizing Flow (CNF) from a simple and easy-to-sample source distribution to an approximation of the posterior. The method has been shown to achieve state-of-the-art performance in several generative as well as inference tasks. We focused on the ability of FMPE to learn a multimodal posterior, showing good performance in a toy example of a mixture of Gaussians and suboptimal performance on a real radiocarbon dating example. Further work should assess a wider variety of path designs at higher computational budgets to better explore the limitations of FMPE.

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1 Introduction

Much of Bayesian inference relies on our ability to use the posterior, which is often a complex intractable distribution. An entire field of scientific research - stochastic simulations - is concerned with obtaining representative samples from such complex distributions. An alternative approach is that of variational inference; however, until recently this alternative has been quite limited and impractical. Latest advancements in variational inference take advantage of modern deep learning methods to offer scalable solutions, achieving state-of-the-art performance and injecting new energy into the field. In this thesis, we shall explore the latest method of Flow Matching for Posterior Estimation and apply it to novel inference problems.

2 Variational Inference

In Bayesian inference, we draw conclusions about the parameters $\theta \in \Theta$ by updating our prior beliefs, $p(\theta)$, to posterior beliefs

$$p(\theta|x_{obs}) = \frac{p(\theta)p(x_{obs}|\theta)}{p(x_{obs})},$$

having observed data $x_{obs} \in \mathcal{X}$. Excluding simplest problems, the posterior is intractable - we cannot compute it analytically. In practice, we instead use a sample from the posterior to approximate any desired expectation (any estimator) over the posterior distribution.

Traditionally, we would obtain an approximate sample from the exact posterior using MCMC sampling techniques. These methods generally enjoy good theoretical convergence guarantees, but can become quite complex and expensive with higher dimensional posteriors [Gelman et al., 1995].

Variational inference instead aims to obtain the exact sample from an approximate posterior. The problem is turned into an optimization where we seek to find the best approximation to the posterior from a parameterized set of easy to sample distributions.

Formally, given a target posterior $p(\theta|x)$, we seek to find a tractable parameterized approximation $q_w(\theta)$, $w \in \mathbb{R}^p$ that minimizes the difference to the posterior. We quantify this difference using a distance measure such as the reverse KL divergence:

$$\mathbb{D}_{\text{KL}}(q_w(\cdot) \parallel p(\cdot|x)) := \mathbb{E}_{\theta \sim q_w(\theta)} \left[\log \frac{q_w(\theta)}{p(\theta|x)} \right] \quad (1)$$

Our approximation is then q_w^* such that

$$q_w^* = \underset{q_w \in \mathcal{Q}}{\operatorname{argmin}} \mathbb{D}_{\text{KL}}(q_w \parallel p) \quad (2)$$

where $\mathcal{Q} = \{q_w | w \in \mathbb{R}^p\}$ is a selected parametric family of distributions over which we optimize. The KL divergence is always non-negative and $\mathbb{D}_{\text{KL}}(q_w \parallel p) = 0$ if and only if $q_w = p$ (up to differences on sets of zero measure).

Expectations from the posterior are then estimated by

$$\begin{aligned}\mathbb{E}_{\theta \sim p(\theta|x)} [g(\theta)] &\approx \mathbb{E}_{\theta \sim q_w^*(\theta)} [g(\theta)] \\ &\approx \frac{1}{n} \sum_{i=1}^n g(\theta_i), \text{ with } \{\theta_i\}_{i=1}^n \sim q_w^*\end{aligned}$$

for any desired function g .

2.0.1 ELBO optimization

In the context of Bayesian inference the posterior is intractable and the KL divergence in the form described above cannot be evaluated. Assuming the posterior can be evaluated up to a constant independent of θ (i.e. is not doubly intractable), we reformulate the KL divergence using the joint distribution $p(\theta, x) = p(\theta)p(x|\theta)$

$$\begin{aligned}\mathbb{D}_{\text{KL}}(q_w(\cdot) \parallel p(\cdot|x)) &= \mathbb{E}_{\theta \sim q_w(\theta)} \left[\log \frac{q_w(\theta)}{p(\theta, x)/p(x)} \right] \\ &= \mathbb{E}_{\theta \sim q_w(\theta)} \left[\log \frac{q_w(\theta)}{p(\theta, x)} \right] + \log p(x) \\ &=: -\mathcal{L}_{\text{ELBO}}(w) + \log p(x)\end{aligned}\tag{3}$$

The ELBO term is the part of the divergence we can optimize. We can see from the last equality that

$$\begin{aligned}\mathcal{L}_{\text{ELBO}}(w) &:= -\mathbb{E}_{\theta \sim q_w(\theta)} \left[\log \frac{q_w(\theta)}{p(\theta, x)} \right] \\ &= \log p(x) - \mathbb{D}_{\text{KL}}(q_w \parallel p) \\ &\leq \log p(x)\end{aligned}$$

since $\mathbb{D}_{\text{KL}} \geq 0$, making $\mathcal{L}_{\text{ELBO}}$ a lower bound for the log model evidence (hence the name). Maximizing $\mathcal{L}_{\text{ELBO}}$ is a way to minimize the KL divergence. We can implement this optimization using stochastic gradient descent, where we employ a reparameterisation trick to express the gradient. Let there be a way to express $\theta \sim q_w(\theta)$ as some invertible deterministic transformation $\theta = f(z; w)$ of a base variable z from some simple (easy to sample) distribution $z \sim r(z)$ independent of w . We can then write the $\mathcal{L}_{\text{ELBO}}$ as follows:

$$\begin{aligned}\mathcal{L}_{\text{ELBO}}(w) &= -\mathbb{E}_{\theta \sim q_w(\theta)} \left[\log \frac{q_w(\theta)}{p(\theta, x)} \right] \\ &= \mathbb{E}_{\theta \sim q_w(\theta)} [\log p(\theta, x) - \log q_w(\theta)] \\ &= \mathbb{E}_{z \sim r(z)} [\log p(f(z; w), x) - \log r(z) + \log (|\det(J_f)|)] \\ &= \mathbb{E}_{z \sim r(z)} [\log p(f(z; w), x) + \log (|\det(J_f)|)] + \mathbb{H}(r)\end{aligned}$$

where J_f is the Jacobian of f and $\mathbb{H}(r)$ is the entropy of r . Taking the gradient $\nabla_w \mathcal{L}_{\text{ELBO}}(w)$ is now straightforward since the expectations are no longer dependent on w

$$\nabla_w \mathcal{L}_{\text{ELBO}}(w) = \mathbb{E}_{z \sim r(z)} [\nabla_w \log p(f(z; w), x) + \nabla_w \log (|\det(J_f)|)]$$

2.0.2 Doubly intractable setting

In doubly intractable settings we cannot evaluate $\nabla_w p(x|f(z; w))$ and thus cannot evaluate $\nabla_w \log p(f(z; w), x)$, making the above approach impossible. Instead, we can take the forward KL divergence

$$\begin{aligned}\mathbb{D}_{\text{KL}}(p(\cdot|x) \parallel q_w(\cdot)) &= \mathbb{E}_{\theta \sim p(\theta|x)} \left[\log \frac{p(\theta|x)}{q_w(\theta)} \right] \\ &= -\mathbb{E}_{\theta \sim p(\theta|x)} [\log q_w(\theta)] - \mathbb{H}(p(\theta|x))\end{aligned}$$

Expectations over the posterior are not easily available (entire goal of VI is to obtain those), so this cannot be used directly as an objective. However, we can instead approximate the joint $p(\theta, x)$ by $q_w(\theta, x)$ at every $x \in \mathcal{X}$ and obtain an approximation to the posterior $q_w(\theta|x_{\text{obs}}) \propto q_w(\theta, x_{\text{obs}})$. We can approximate expectations over the joint by sampling it like $(\theta, x) \sim p(\theta)p(x|\theta)$, so the objective

$$\mathcal{L}(w) := -\mathbb{E}_{(\theta, x) \sim p(\theta, x)} [\log q_w(\theta)]$$

is tractable and optimizable. This is a more expensive optimization problem since we optimize for all $x \in \mathcal{X}$, instead of just at x_{obs} . That is to say, we amortise. On the other hand, if we want to evaluate the posterior at different values of x , we still optimize only once.

To achieve perfect fit, it is necessary for the posterior to be within the parameterized family of distributions $\mathcal{Q} = \{q_w | w \in \mathbb{R}^p\}$ over which we optimize, which is generally not the case. It is thus important to understand how inaccuracies manifest when perfect fit is not possible. The reverse KL divergence is infinite if $p(\theta|x) = 0$ while $q(\theta) > 0$ for some θ ¹. This forces the solution to never extend beyond the support of the posterior and typically leads optimal q_w to be an overconfident low variance approximation that underestimates the support. Conversely, the forward KL divergence is infinite if $q(\theta) = 0$ while $p(\theta|x) > 0$. This forces the solution to cover the entire support of the posterior, typically leading the optimal q_w to be an under-confident high variance approximation with a mass covering property.

The quality of a VI posterior approximation is limited by the expressiveness of the family of distributions over which we optimize, the ability of the objective function to distinguish a good approximation, and the ability of the optimization algorithm to find a good optimum of the objective. The difficulty of finding a good VI approximation is thus the difficulty of finding a good optimum among a broad enough set of possible solutions.

2.0.3 Evolution of Variational Inference methods

The classical example of a variational Bayes method is the Coordinate Ascent (CAVI) algorithm. CAVI uses the mean-field variational family $\mathcal{Q} = \{q(\theta) = \prod_{i=1}^p q_i(\theta_i)\}$ where all factors are mutually independent and can thus be updated iteratively. We iteratively update across factors, each update increasing $\mathcal{L}_{\text{ELBO}}$ until we eventually plateau at a local

¹Technically at $p(\theta|x) = 0$ the divergence is undefined since we divide by zero. Practically we say it is infinite since the relevant real scenario we consider is the behavior in the far tails of $p(\theta|x)$ where the posterior density is very nearly zero.

optimum [Blei et al., 2017]. While simple, this algorithm is limited by the size of Q . Crucially, with such a family we can never capture any between-factor correlations.

Until the developments of the last decade, variational inference has mostly been restricted to the mean-field family. The recent advancements can be summarized by two broad directions. First, the inclusion of deep learning methods for scalable learning of the optimal approximation. Second, the concurrent evolution of ever-broader parameterized families of distributions over which we can efficiently optimize.

The Variational Auto-Encoder (VAE) introduced and popularized a neural network parameterization of the posterior for local (per data point) latent variables [Kingma, 2013]. Optimized using the ELBO objective, the original VAE used a simple multivariate Gaussian with a diagonal covariance matrix as the parameterized family of distributions. Salimans et al. [2015] applied deep learning directly to variational inference for global parameters and explored connections to classical MCMC methods. The strongest limitation became the size of the family Q . Rezende and Mohamed [2015] introduced Normalizing Flows (NF), a family parameterized by a series of invertible transformations of a simple base distribution, designed such that the $\nabla_w \log(|\det(J_f)|)$ part of the ELBO gradient was computationally inexpensive. Arbitrary precision could be achieved by scaling the number of transformations, creating a much broader family over which optimization was possible. Louizos and Welling [2017] showed how to apply Normalizing Flows to Bayesian neural networks. A summary of the research on NFs for probabilistic modelling and inference is provided by Papamakarios et al. [2021]. Most recently, Normalizing Flows were superseded by Continuous Normalizing Flows (CNF) [Chen et al., 2018], where discrete transformation were replaced by a continuous evolution. Impressively, moving to a more general continuous setting allowed for an elegant theoretical description of the problem with an ordinary differential equation (ODE). Finally, Flow Matching (FM) replaced the ELBO as a new objective that allows the training of generative models with CNFs that circumvents the calculation of the relatively expensive ODEs during optimization [Lipman et al., 2022]. Flow Matching for Posterior Estimation (FMPE) is then the application of this method for variational inference [Wildberger et al., 2024].

In summary, variational inference as a field has recently seen substantial improvements, fueled by the ability of scalable deep learning models to efficiently learn optimal approximations to the posterior from new, vastly more expressive families of proposal distributions. In this thesis, we will focus on FMPE as the latest technique for variational inference. The following sections will describe the Continuous Normalizing Flows family of distributions, the Flow Matching objective, and the reformulation of the objective for posterior estimation.

3 Methodology

3.1 Continuous Normalizing Flows

In this section we present Continuous Normalizing Flows (CNF) [Chen et al., 2018], a method of generating a complex distribution from a simple one using flows. In the framework of variational inference, CNFs serve to create an easy-to-sample yet broad parameterized family of proposal distributions from which we seek the optimal approximation to the posterior.

Starting with a source distribution that we can easily sample (e.g. Gaussian noise), we can define time-dependent diffeomorphic map that transforms the source into a more complex distribution. The mapping is defined through instantaneous rates of change in space with regards to time, in effect creating a continuous flow in time from the source to end distribution. Sampling the end distribution is then as simple as sampling the source distribution and applying the deterministic transformation.

Considering the space of the target $\Theta = \mathbb{R}^d$, let $u : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a time-dependent vector field, with u_t denoting the partial evaluation $u(t, \cdot)$ of u at time t . We use u to define a continuous normalizing flow $\phi : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ through the following ODE:

$$\frac{d}{dt}\phi_t(\theta) := u_t(\phi_t(\theta)) \quad (4)$$

$$\phi_0(\theta) = \theta \quad (5)$$

In other words, $\phi_t(\theta)$ describes the position of a particle $\theta \in \mathbb{R}^d$ after following the trajectory ϕ up to time t . This trajectory is uniquely defined through setting the derivative $\frac{d}{dt}\phi_t(\theta)$ at every point in time. We say that u generates q_1 (from q_0).

We can now use this flow to obtain a time-dependent probability density $q : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}_{>0}$. Starting at a simple source density q_0 , we obtain q_t through the push-forward operation $q_t = [\phi_t]_* q_0$, which simply applies the standard multivariate change-of-variable formula for the ϕ_t transformation

$$[\phi_t]_* q_0(\theta) := q_0(\phi_t^{-1}(\theta)) \det \left[\frac{\partial \phi_t^{-1}}{\partial \theta}(\theta) \right] \quad (6)$$

Importantly, q_t is a valid density ($\int q_t(\theta) d\theta = 1, q_t(\theta) \geq 0 \forall \theta \in \mathbb{R}^d$) and u_t generates q_t if and only if they satisfy the continuity equation:

$$\frac{d}{dt}q_t(\theta) + \nabla_\theta \cdot (q_t(\theta)u_t(\theta)) = 0, \quad (7)$$

where ∇_θ is the divergence operator with respect to θ [Villani et al., 2009].

The CNF model allows us to obtain wide range of complex distributions by specifying the vector fields u_t . We can model these fields using a neural network with learnable parameters $w \in \mathbb{R}^p$. This allows a scalable architecture that can be adjusted to the complexity of the task at hand. Originally proposed by [Chen et al., 2018], CNFs were trained using maximum likelihood methods, requiring expensive numerical ODE solving during training. In the following section we present the Flow Matching objective which circumvents this requirement and provides an effective approach for training the CNF model [Lipman et al., 2022]

3.2 Flow Matching objective

[Lipman et al., 2022] propose the Flow Matching objective as a simulation-free and general objective for training CNFs. In the framework of variational inference, Flow Matching is the objective helping us select the best approximation to the posterior from the CNF parameterized family of proposals. Before delving into posterior estimation, we first present Flow Matching in the generative case, where the CNF model is trained to create an approximation to a generic target distribution p .

3.2.1 Generative Flow Matching

Let us denote the unknown target distribution as p . We wish to approximate p by q_1 , a density specified by a CNF model with probability densities $(q_t)_{t \in [0,1]}$ and corresponding vector fields $(u_t)_{t \in [0,1]}$. Supposing we know q_t and u_t , we can train a neural network $v : [0, 1] \times \mathbb{R}^d \times \mathbb{R}^p \rightarrow \mathbb{R}^d$ to approximate u with a simple regression objective:

$$\mathcal{L}_{\text{FM}}(w) := \mathbb{E}_{t \sim \mathcal{U}[0,1], \theta \sim q_t(\theta)} [\|v_t(\theta; w) - u_t(\theta)\|^2] \quad (8)$$

We can apply standard deep learning methods to train the weights w of the neural network, using $\mathcal{L}_{\text{FM}}$ as the loss to be minimized. At zero loss, the learned vector field v generates q_1 exactly.

While this objective achieves the desired optimum, it is intractable; suitable q_t and u_t are typically unknown, for any given q_0 there are many options of paths such that $q_1 \approx p$, there are multiple viable approximations q_1 , and how to obtain appropriate u_t to generate q_t is generally unclear. Lipman et al. [2022] solve all these problems by proposing a *per sample* conditional construction of probability densities and vector fields that yields the same optimum of $u_t(\theta)$ for $v_t(\theta; w)$ under a new loss $\mathcal{L}_{\text{CFM}}$.

3.2.2 Conditional Flow Matching

Denoting $\eta \sim p$ to be a sample observation from the unknown target distribution p , Lipman et al. [2022] define a conditional probability densities $q_t(\theta|\eta)$ with corresponding conditional vector fields $u_t(\theta|\eta)$ that generate them. Let the design be such that $q_0(\theta|\eta) = q(\theta)$ so the source distribution is independent of the conditioning, and such that $q_1(\theta|\eta)$ is concentrated around η , for example $q_1(\theta|\eta) = \mathcal{N}(\theta|\eta, \sigma_{\min}^2 I_d)$ for some small $\sigma_{\min} > 0$. It is useful to think of this as a "noisy Dirac delta" function around the observation η . The relationship between the conditional and marginal probability density and vector field is defined as follows:

$$q_t(\theta) := \mathbb{E}_{\eta \sim p(\eta)} [q_t(\theta|\eta)] \quad (9)$$

$$\begin{aligned} &= \int q_t(\theta|\eta) p(\eta) d\eta \\ u_t(\theta) &:= \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [u_t(\theta|\eta)] \quad (10) \\ &= \int u_t(\theta|\eta) \frac{q_t(\theta|\eta) p(\eta)}{q_t(\theta)} d\eta \end{aligned}$$

Put in words, marginalizing $q_t(\theta|\eta)$ over $p(\eta)$ will give the target probability density and marginalizing $u_t(\theta|\eta)$ over $q_t(\eta|\theta)$ will give the vector field $u_t(\theta)$. We can see how q_1

approximates p since²

$$\begin{aligned} q_1(\theta) &= \int q_1(\theta|\eta)p(\eta) \, d\eta \\ &\approx \int \delta_\theta(\eta)p(\eta) \, d\eta \\ &= p(\theta) \end{aligned}$$

It is not immediately clear from this definition that such a vector field u_t will generate the desired probability density q_t . Lipman et al. [2022] prove this in the following theorem:

Theorem 3.1. *Given the conditional vector field $u_t(\theta|\eta)$ that generates the conditional probability density $q_t(\theta|\eta)$, the marginal vector field u_t given in equation 10 generates the marginal probability density q_t given in equation 9.*

Proof. We show that q_t and u_t satisfy the continuity equation (7):

$$\begin{aligned} \frac{d}{dt}q_t(\theta) &\stackrel{\heartsuit}{=} \int \left(\frac{d}{dt}q_t(\theta|\eta) \right) p(\eta) \, d\eta \\ &\stackrel{\text{eq. (7)}}{=} - \int \nabla_\theta \cdot (u_t(\theta|\eta)q_t(\theta|\eta))p(\eta) \, d\eta \\ &\stackrel{\heartsuit}{=} - \nabla_\theta \cdot \left(\int u_t(\theta|\eta)q_t(\theta|\eta)p(\eta) \, d\eta \right) \\ &\stackrel{\text{eq. (10)}}{=} - \nabla_\theta \cdot (u_t(\theta)q_t(\theta)) \end{aligned}$$

where steps $\heartsuit$ assume the integrand and its corresponding derivative are continuous, so that we can differentiate under the integral sign (Leibniz rule)³.

□

We can now to specify a tractable objective $\mathcal{L}_{\text{CFM}}(w)$ using only the conditional probability densities and vector fields such that the optimum $u_t(\theta)$ is preserved.

$$\operatorname{argmin}_{v_t(\theta;w)} \mathcal{L}_{\text{FM}}(w) = \operatorname{argmin}_{v_t(\theta;w)} \mathcal{L}_{\text{CFM}}(w)$$

Let the Conditional Flow Matching objective be:

$$\mathcal{L}_{\text{CFM}}(w) = \mathbb{E}_{t \sim \mathcal{U}[0,1], \eta \sim p(\eta), \theta \sim q_t(\theta|\eta)} [\|v_t(\theta; w) - u_t(\theta|\eta)\|^2] \quad (11)$$

Lipman et al. [2022] show this objective achieves the same optimum as $\mathcal{L}_{\text{FM}}$ up to a constant independent of w by proving the equality of the gradients $\nabla_w \mathcal{L}_{\text{FM}}(w) = \nabla_w \mathcal{L}_{\text{CFM}}(w)$. Here, we use our own argument to directly show that $\mathcal{L}_{\text{CFM}}$ is minimized at $\hat{v}_t(\theta; w) = u_t(\theta)$.

Theorem 3.2. *The $\mathcal{L}_{\text{CFM}}(w)$ loss is minimized by $\hat{v}_t(\theta; w) = u_t(\theta)$*

²In the limit $\sigma_{\min} \downarrow 0$, the approximation turns exact.

³To be exact, for the first equality we assume both $q_t(\theta|\eta)p(\eta)$ and $\left(\frac{d}{dt}q_t(\theta|\eta)\right)p(\eta)$ are continuous in (t, η) . For the third equality, we assume both $u_t(\theta|\eta)q_t(\theta|\eta)p(\eta)$ and $\nabla_\theta \cdot (u_t(\theta|\eta)q_t(\theta|\eta))p(\eta)$ are continuous in (θ, η) .

Proof. Using Bayes theorem we replace sampling the joint (η, θ) from the prior $\eta \sim p(\eta), \theta \sim q_t(\theta|\eta)$ by sampling from the posterior $\eta \sim q_t(\eta|\theta), \theta \sim q_t(\theta)$

$$\begin{aligned}\mathcal{L}_{\text{CFM}}(w) &= \mathbb{E}_{t \sim \mathcal{U}[0,1], \eta \sim p(\eta), \theta \sim q_t(\theta|\eta)} [\|v_t(\theta; w) - u_t(\theta|\eta)\|^2] \\ &= \mathbb{E}_{t \sim \mathcal{U}[0,1], \theta \sim q_t(\theta)} \left[\mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|v_t(\theta; w) - u_t(\theta|\eta)\|^2] \right]\end{aligned}\quad (12)$$

The inner expectation can be written as

$$\begin{aligned}\mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|v_t(\theta; w) - u_t(\theta|\eta)\|^2] &= \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|v_t(\theta; w) - u_t(\theta) + u_t(\theta) - u_t(\theta|\eta)\|^2] \\ &= \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|v_t(\theta; w) - u_t(\theta)\|^2] \\ &\quad + 2\mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\langle v_t(\theta; w) - u_t(\theta), u_t(\theta) - u_t(\theta|\eta) \rangle] \\ &\quad + \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|u_t(\theta) - u_t(\theta|\eta)\|^2] \\ &= \|v_t(\theta; w) - u_t(\theta)\|^2 \\ &\quad + 2\langle v_t(\theta; w) - u_t(\theta), u_t(\theta) - \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [u_t(\theta|\eta)] \rangle \\ &\quad + \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|u_t(\theta) - u_t(\theta|\eta)\|^2] \\ &\stackrel{\heartsuit}{=} \|v_t(\theta; w) - u_t(\theta)\|^2 \\ &\quad + \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|u_t(\theta) - u_t(\theta|\eta)\|^2] \\ &\propto \|v_t(\theta; w) - u_t(\theta)\|^2\end{aligned}$$

with proportionality being up to an additive constant with regards to the optimization parameters w . In step $\heartsuit$ we applied the definition of the marginal vector field (10) to cancel the second term. Plugging the result back into equation 12, we obtain:

$$\begin{aligned}\mathcal{L}_{\text{CFM}} &= \mathbb{E}_{t \sim \mathcal{U}[0,1], \theta \sim q_t(\theta)} \left[\mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|v_t(\theta; w) - u_t(\theta|\eta)\|^2] \right] \\ &\propto \mathbb{E}_{t \sim \mathcal{U}[0,1], \theta \sim q_t(\theta)} [\|v_t(\theta; w) - u_t(\theta)\|^2] \\ &= \mathcal{L}_{\text{FM}}\end{aligned}$$

where $\mathcal{L}_{\text{FM}}$, a classical regression objective, is clearly minimized by $\hat{v}_t(\theta; w) = u_t(\theta)$. Since the proportionality maintains exactly the terms dependent on the optimized parameters w , the minimizer of $\mathcal{L}_{\text{CFM}}$ is identical ⁴.

Alternatively, in equation 12 we could have recognized $\mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|v_t(\theta; w) - u_t(\theta|\eta)\|^2]$ to be the expected posterior loss (posterior $\eta \sim q_t(\eta|\theta)$) of the decision rule $v_t(\theta; w)$, estimating $u_t(\theta|\eta)$ (as a function of the parameter η), under the quadratic loss function. By standard decision theory results (p. 228, Lehmann and Casella [2006]), the expected posterior loss is minimized by the Bayes rule, which for the quadratic loss is the posterior mean

$$\begin{aligned}\hat{v}_t(\theta; w) &= \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [u_t(\theta|\eta)] \\ &= u_t(\theta).\end{aligned}$$

□

⁴While the optimizer is identical, the optimum of $\mathcal{L}_{\text{CFM}}$ and $\mathcal{L}_{\text{FM}}$ differs by a constant $C = -\|u_t(\theta)\| + \mathbb{E}_{\eta \sim q_t(\eta|\theta)} [\|u_t(\theta|\eta)\|^2]$ independent of w .

The $\mathcal{L}_{\text{CFM}}$ loss is a simulation-free objective that circumvents the need to access the intractable marginal probability density or vector field. Additionally, we have substantial freedom in our design choice for the conditional densities and fields. The following section will discuss common options.

3.2.3 Conditional probability densities and vector fields

The $\mathcal{L}_{\text{CFM}}$ objective does not restrict our choice of conditional probability densities and vector fields. Lipman et al. [2022] propose conditional probability densities of a Gaussian form:

$$\begin{aligned} q_t(\theta|\eta) &= \mathcal{N}(\theta|\mu_t(\eta), \sigma_t(\eta)^2 I_d) \\ \mu &: [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d \\ \sigma &: [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}_{>0} \\ \mu_0(\eta) &= 0, \mu_1(\eta) = \eta \\ \sigma_0(\eta) &= 1, \sigma_1(\eta) = \sigma_{\min} \end{aligned} \tag{13}$$

for functions $\mu(\eta), \sigma(\eta)$ of our choosing. Since setting a probability density does not fix a unique vector field that can generate it, we must also choose an appropriate vector field. Seeking to avoid rotational transformations to which the Gaussian distribution is invariant, we desire a conditional flow corresponding to the canonical transformation of a Gaussian distribution

$$\varphi_t(\theta|\eta) := \sigma_t(\eta)\theta + \mu_t(\eta) \tag{14}$$

This fixes a unique conditional vector field.

Theorem 3.3. *Let $q_t(\theta|\eta)$ be a Gaussian probability density as specified in 13 and $\varphi_t(\theta|\eta)$ its corresponding flow map as specified in 14. Then the unique vector field that defines $\varphi_t(\theta|\eta)$ has the form*

$$u_t(\theta|\eta) = \frac{\sigma'_t(\eta)}{\sigma_t(\eta)}(\theta - \mu_t(\eta)) + \mu'_t(\eta) \tag{15}$$

where $\sigma'_t(\eta), \mu'_t(\eta)$ are derivatives with respect to t (Thm. 3 Lipman et al. [2022]).

Proof. The inverse of $\varphi_t(\theta|\eta)$ is⁵,

$$\varphi_t^{-1}(\theta|\eta) = \frac{\theta - \mu_t(\eta)}{\sigma_t(\eta)}$$

and the derivative of the flow with respect to t is

$$\frac{d}{dt}\varphi_t(\theta|\eta) = \sigma'_t(\eta)\theta + \mu'_t(\eta)$$

⁵The inverse exists since flow is diffeomorphic [Lipman et al., 2022].

Plugging both into eq. 4 from the definition of a CNF, we obtain:

$$\begin{aligned}
& \frac{d}{dt} \varphi_t(\theta|\eta) = u_t(\varphi_t(\theta|\eta)|\eta) \\
\Rightarrow & \frac{d}{dt} \varphi_t(\varphi_t^{-1}(\theta|\eta)|\eta) = u_t(\varphi_t(\varphi_t^{-1}(\theta|\eta)|\eta)|\eta) \\
\Leftrightarrow & \frac{d}{dt} \varphi_t \left(\frac{\theta - \mu_t(\eta)}{\sigma_t(\eta)} \middle| \eta \right) = u_t(\theta|\eta) \\
\Leftrightarrow & \sigma_t'(\eta) \frac{\theta - \mu_t(\eta)}{\sigma_t(\eta)} + \mu_t'(\eta) = u_t(\theta|\eta)
\end{aligned}$$

□

We are free to choose any differentiable functions for $\mu_t(\eta)$ and $\sigma_t(\eta)$, subject to the above specified boundary conditions (behavior fixed at $t = 0, t = 1$). Lipman et al. [2022] list the following (non-exhaustive) options:

- **reversed (noise→data) Variance Exploding diffusion path** [Song et al., 2020]

$$\begin{aligned}
& \mu_t(\eta) = \eta \\
& \sigma_t(\eta) = \sigma_{1-t}, \sigma_0 = 0, \sigma_1 \gg 1, \sigma_t \text{ increasing} \\
\Rightarrow & q_t(\theta|\eta) = \mathcal{N}(\theta|\eta, \sigma_{1-t}^2 I) \\
\Rightarrow & u_t(\theta|\eta) = \frac{\sigma_{1-t}'}{\sigma_{1-t}}(\theta - \eta)
\end{aligned}$$

- **reversed (noise→data) Variance Preserving diffusion path** [Song et al., 2020]

$$\begin{aligned}
& \mu_t(\eta) = \alpha_{1-t}\eta \\
& \sigma_t(\eta) = \sqrt{1 - \alpha_{1-t}^2} \\
& \alpha_t = e^{-\frac{1}{2}T(t)} \\
& T(t) = \int_0^t \beta(s) ds
\end{aligned}$$

where β is the noise scale function. Equations 13 and 15 give the corresponding conditional probability density and vector field, respectively.

- **conditional Optimal Transport (cOT)**

$$\begin{aligned}
& \mu_t(\eta) = t\eta \\
& \sigma_t(\eta) = 1 - (1 - \sigma_{\min})t \\
\Rightarrow & \varphi_t(\theta|\eta) = (1 - (1 - \sigma_{\min})t)\theta + t\eta
\end{aligned}$$

Here, we let the mean to linearly interpolate in time $0 \rightarrow \eta$ and the standard deviation to linearly interpolate $1 \rightarrow \sigma_{\min}$. Lipman et al. [2022] show the corresponding conditional flow $\varphi_t(\theta|\eta)$ is the Optimal Transport displacement map between $q_0(\theta)$ and $q_1(\theta|\eta)$. The appeal of this design is that the direction and speed of the conditional vector fields are constant in time, suggesting an easier fit for a parametric model. They confirm this experimentally. Importantly, we must remark that an optimal conditional flow does not imply the *marginal* vector field to be an optimal transport solution.

3.2.4 Improved designs

Most recently, Tong et al. [2023] have significantly improved performance of Flow-Matching and similar algorithms by building up the theory for minibatch marginal optimal transport solutions. They generalize the designs of conditional probability densities and vector fields as follows:

Independent coupling

Conditioning on a target point η can be generalized to conditioning on a source point η_0 as well as target point η_1 and creating an independent coupling $q(\eta_0, \eta_1) = q_0(\eta_0)p(\eta_1)$ of source and target distributions, respectively. Previous designs all required a Gaussian noise as a source distribution - this restriction is here lifted. For a Gaussian flow as given in 14, Tong et al. [2023] specify

$$\begin{aligned}\mu_t(\eta_0, \eta_1) &= t\eta_1 + (1-t)\eta_0 \\ \sigma_t(\eta_0, \eta_1) &= \sigma \\ \Rightarrow q_t(\theta|\eta_0, \eta_1) &= \mathcal{N}(\theta|t\eta_1 + (1-t)\eta_0, \sigma^2 I) \\ \Rightarrow u_t(\theta|\eta_0, \eta_1) &= \eta_1 - \eta_0\end{aligned}\tag{16}$$

We are still free to specify a non-Gaussian $q_0(\eta_0)$. Setting $q_0(\eta_0) = \mathcal{N}(\eta_0|0, I)$ and $\sigma_t^2(\eta_0, \eta_1) = (\sigma_{\min}t)^2 + 2\sigma_{\min}t(1-t)$ will recover the conditional OT design (see Prop. B.1 Tong et al. [2023]).

Marginal optimal transport (OT)

It is desirable to find the simplest and most straightforward possible way to reach the target distribution. Optimal transport theory is the formalization of this question: the dynamic optimal transport problem seeks a continuous evolution of one distribution to another that minimizes a total transport cost over time. Using the 2-Wasserstein distance with respect to the Euclidean distance cost $c(\theta, \rho) = \|\theta - \rho\|$ between distributions q_0 and p on $\mathbb{R}^d$, the dynamic form of the problem is

$$W(q_0, p)_2^2 = \inf_{q_t, u_t} \int_{\mathbb{R}^d} \int_0^1 q_t(\theta) \|u_t(\theta)\|^2 dt d\theta, \tag{18}$$

with boundary conditions $q_1 = p$, q_0 fixed and q_t, u_t satisfying the continuity equation (7). This specification uses vector fields u_t to generate q_t , just like the CNF model. Finding the dynamic optimal transport solution would yield the cheapest (most direct) CNF model to transform our chosen, easy-to-sample q_0 into the desired target p .

Finding the dynamic OT minimum and teaching it to our parameterized model v_t is not trivial. First, we use the equivalence of the dynamic and static optimal transport formulation (assuming q_0, p are compactly supported distributions with bounded density. Proven by Benamou and Brenier [2000]) to change the optimization into the static form:

$$W(q_0, p)_2^2 = \inf_{\pi \in \Pi} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(\theta, \rho)^2 d\pi(\theta, \rho) \tag{19}$$

where Π is the set of all joint probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with q_0, p marginals. Assuming we know how to solve this optimization, we can use $q(\eta_0, \eta_1) := \operatorname{arginf}_{\pi \in \Pi} W(q_0, p)_2^2$

to jointly sample (η_0, η_1) and use these samples in optimization of $\mathcal{L}_{\text{CFM}}$. In their proposition 3.4, Tong et al. [2023] show that $q(\eta_0, \eta_1)$ can indeed be applied to train the $\mathcal{L}_{\text{CFM}}$ objective, and additionally, that with the conditional probability densities and vector fields as in equations 16 and 17, respectively, the vector field u_t solves the dynamic optimal transport problem between q_0 and p .

Finally, we must compute the optimal transport plan π that solves the static OT problem. This is a problem of considerable computational complexity - cubic time and quadratic memory complexity in the number of samples [Tong et al., 2020]. For larger samples, we instead use a minibatch OT approximation where the optimal transport problem is only solved for a batch of the data at a time [Fratras et al., 2021]. Tong et al. [2023] show state-of-the-art performance of this approach. For computational reasons, we do not attempt this design here.

3.2.5 Flow Matching for Posterior estimation (FMPE)

The Flow Matching objective can be adapted from a generative model to a variational Bayesian inference technique for posterior estimation [Wildberger et al., 2024]. Neural posterior estimation (NPE) trains a neural network to directly approximate the true posterior $p(\theta|x)$ using the forward KL divergence $\mathcal{L}_{\text{NPE}} = -\mathbb{E}_{p(\theta)p(x|\theta)} [\log q(\theta|x)]$. Exchanging the objective for a Flow Matching loss will allow us to circumvent the evaluation of the estimated log-density $\log q(\theta|x)$, which in the case of CNFs involves solving an ODE. This improvement is a direct analogy to the advances Flow Matching has made in the generative case. Classical NPE is a simulation based inference (SBI) technique with competitive performance across tasks [Lueckmann et al., 2021], with the benefit of being able to amortize across observations and to evaluate the approximated density. Additionally, the flexibility of neural network architectures allows us to apply NPE to large scale inference problems beyond the reach of more traditional methods [Greenberg et al., 2019]. Wildberger et al. [2024] show how Flow Matching preserves these properties while demonstrating state-of-the-art performance on select inference problems.

To construct the $\mathcal{L}_{\text{FMPE}}$ objective, we simply replace the conditioning on the target distribution $p(\eta)$ in $\mathcal{L}_{\text{CFM}}$ with conditioning on the joint distribution $(\eta, x) \sim p(\eta)p(x|\eta)$ and we construct the conditional probability density and vector field only in the space of θ :

$$\mathcal{L}_{\text{FMPE}} = \mathbb{E}_{t \sim q(t), \eta \sim p(\eta), x \sim p(x|\eta), \theta \sim q_t(\theta|\eta)} [\|v_{t,x}(\theta; w) - u_t(\theta|\eta)\|^2] \quad (20)$$

The conditioning on $\eta \in \mathbb{R}^d$ is directly in $u_t(\theta|\eta)$, whereas $x \in \mathbb{R}^m$ is taken as input only to the neural network $v(\cdot; w) : [0, 1] \times \mathbb{R}^{d+m} \times \mathbb{R}^p \rightarrow \mathbb{R}^d$ and the conditioning is implicit by taking the expectation with respect to x . Additionally, the sampling of t need not be uniform, allowing more training capacity to be assigned to times for which the vector field is more complex. Wildberger et al. [2024] suggest a power-law distribution $q_\alpha(t) \propto t^{1/(1+\alpha)}$, $\alpha > 0$ giving more importance to larger values of t .

The proof that $\mathcal{L}_{\text{FMPE}}$ is minimized by $\widehat{v}_{t,x}(\theta; w) = u_t(\theta|\eta)$ and that this optimum generates the posterior is a direct analogy to the proof of Theorem 3.2. Nevertheless, we state it here, since to the best of our knowledge, no paper has explicitly given it.

Theorem 3.4. *The $\mathcal{L}_{\text{FMPE}}$ objective as given in equation 20 is minimized by $\widehat{v}_{t,x}(\theta; w) = u_{t,x}(\theta)$. The learned approximation $q_{1,x}(\theta)$ generated by optimal $\widehat{v}_{t,x}(\theta; w)$ then approximates the posterior $p(\theta|x)$.*

Proof. We begin by making explicit the relationship between marginal and conditional probability densities and vector fields:

$$\begin{aligned} q_{t,x}(\theta) &= \mathbb{E}_{\eta \sim p(\eta|x)} [q_t(\theta|\eta)] \\ u_{t,x}(\theta) &= \mathbb{E}_{\eta \sim q_{t,x}(\eta|\theta)} [u_t(\theta|\eta)] \end{aligned} \quad (21)$$

$$q_{t,x}(\eta|\theta) = \frac{q_t(\theta|\eta)p(\eta|x)}{q_{t,x}(\theta)}, \quad (22)$$

where $u_{t,x}(\theta)$ generates $q_{t,x}(\theta)$ (proof identical to that of Theorem 3.1). As in the proof of Theorem 3.2, we now express the objective $\mathcal{L}_{\text{FMPE}}$ in proportion to the optimization parameters w

$$\begin{aligned} \mathcal{L}_{\text{FMPE}}(w) &= \mathbb{E}_{t \sim q(t), (\eta, x) \sim p(\eta)p(x|\eta), \theta \sim q_t(\theta|\eta)} [\|v_{t,x}(\theta; w) - u_t(\theta|\eta)\|^2] \\ &\stackrel{(1)}{=} \mathbb{E}_{t \sim q(t), (\eta, x) \sim p(x)p(\eta|x), \theta \sim q_t(\theta|\eta)} [\|v_{t,x}(\theta; w) - u_t(\theta|\eta)\|^2] \\ &\quad + 2\mathbb{E}_{t \sim q(t), (\eta, x) \sim p(x)p(\eta|x), \theta \sim q_t(\theta|\eta)} [\langle v_{t,x}(\theta; w) - u_{t,x}(\theta), u_{t,x}(\theta) - u_t(\theta|\eta) \rangle] \\ &\quad + \mathbb{E}_{t \sim q(t), (\eta, x) \sim p(x)p(\eta|x), \theta \sim q_t(\theta|\eta)} [\|u_{t,x}(\theta) - u_t(\theta|\eta)\|^2] \\ &\stackrel{(2)}{\propto} \mathbb{E}_{t \sim q(t), x \sim p(x), \eta \sim q_{t,x}(\eta|\theta), \theta \sim q_{t,x}(\theta)} [\|v_{t,x}(\theta; w) - u_{t,x}(\theta)\|^2] \\ &\quad + 2\mathbb{E}_{t \sim q(t), x \sim p(x), \theta \sim q_{t,x}(\theta)} [\langle v_{t,x}(\theta; w) - u_{t,x}(\theta), u_{t,x}(\theta) - \mathbb{E}_{\eta \sim q_{t,x}(\eta|\theta)} [u_t(\theta|\eta)] \rangle] \\ &\stackrel{(3)}{=} \mathbb{E}_{t \sim q(t), x \sim p(x), \theta \sim q_{t,x}(\theta)} [\|v_{t,x}(\theta; w) - u_{t,x}(\theta)\|^2] \end{aligned}$$

In step (1) we use Bayes rule to interchange $(\eta, x) \sim p(\eta)p(x|\eta) \rightarrow (\eta, x) \sim p(x)p(\eta|x)$. In step (2) we remove the last expectation because it is independent of the optimized parameters w (w.r.t. which we take proportionality to), we apply Bayes rule from the definition of the conditional vector field (eq. 22) to interchange $(\eta, \theta) \sim p(\eta|x)q_t(\theta|\eta) \rightarrow (\eta, \theta) \sim q_{t,x}(\eta, \theta)q_{t,x}(\theta)$, and lastly, we use the linearity of the inner product to apply the expectations with regards to η directly to the elements of the inner product. In step (3) we simply recognize the definition of the marginal vector field (eq. 21) to cancel out the second term.

From the last expression we can clearly see that $\mathcal{L}_{\text{FMPE}}$ is minimized by setting $\widehat{v}_{t,x}(\theta; w) = u_{t,x}(\theta)$ point-wise at every point (t, x, θ) . Since $u_{t,x}(\theta)$ generates $q_{1,x}(\theta)$, we have thus shown that optimal $\widehat{v}_{t,x}(\theta; w)$ generates $q_{1,x}(\theta)$. This, in turn, is an approximation of $p(\theta|x)$ since we design the conditional density such that $q_1(\theta|\eta) = \mathcal{N}(\theta|\eta, \sigma_{\min}^2 I_d)$, just like in the generative case

$$\begin{aligned} q_{1,x}(\theta) &= \int q_1(\theta|\eta)p(\eta|x) d\eta \\ &\approx \int \delta_\theta(\eta)p(\eta|x) d\eta \\ &= p(\theta|x) \end{aligned}$$

□

Algorithm 1 Flow Matching for Posterior Estimation

Input: dataset from target $p(\eta)p(x|\eta)$,
time distribution $q(t)$,
conditional probability density and vector field $q_t(\theta|\eta), u_t(\theta|\eta)$,
initial network v .

- 1: **while** Training **do**
- 2: $(\eta, x) \sim p(\eta)p(x|\eta)$ ▷ Conditioning sample from target joint
- 3: $t \sim q(t)$
- 4: $\theta \sim q_t(\theta|\eta)$
- 5: $\mathcal{L}_{\text{FMPE}}(w) \leftarrow \|v_{t,x}(\theta; w) - u_t(\theta|\eta)\|^2$
- 6: $w \leftarrow \text{Optimizer_Update}(w, \nabla_w \mathcal{L}_{\text{FMPE}}(w))$
- 7: **end while**
- 8: **return** Trained v

Wildberger et al. [2024] also derive a theoretical upper bound on the forward KL divergence between the target posterior $p(\theta|x)$ and the learned FMPE approximation $q_{1,x}(\theta)$.

Theorem 3.5. *Let $p_0 = q_0$ and let u_t and v_t be vector fields that generate $p(\theta|x)$ and $q_{1,x}(\theta)$, respectively (from p_0 and q_0 , in the sense exactly described in section 3.1). Assume that p_0 is square integrable and satisfies $|\nabla \ln p_0(\theta)| \leq c(1 + |\theta|)$ and that u_t and v_t have bounded second derivatives. Then there is a constant $C > 0$ such that (for $\text{MSE}_p(u, v) < 1$)*

$$\mathbb{D}_{\text{KL}}(p(\cdot|x) \parallel q_{1,x}(\cdot)) \leq C \text{MSE}_{p(\theta|x)}(u, v)^{\frac{1}{2}}. \quad (23)$$

The proof can be found in the appendix of Wildberger et al. [2024]. While the regularity conditions cannot be guaranteed for general neural network parameterizations of v_t , they provide confidence that FMPE solutions will emphasize mass coverage properties that come with optimizing the forward KL divergence. This notion has been reinforced by empirical results [Wildberger et al., 2024].

Lastly, Wildberger et al. [2024] also derive a parameterization of v_t such that FMPE is forced to learn a Gaussian distribution $q(\theta) = \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ with learned $\hat{\mu}, \hat{\sigma}^2$.

$$v_t(\theta) = \frac{(\sigma_t^2 + (t\hat{\sigma})^2 - \sigma_t)\theta_t + t\hat{\mu}\sigma_t}{t(\sigma_t^2 + (t\hat{\sigma})^2)}, \quad (24)$$

where $\sigma_t = 1 - (1 - \sigma_{\min})t$ is the conditional optimal transport path (so also $\mu_t(\eta) = t\eta$).

4 Experiments

Implementations of the experiments are published on [Github](#).

4.1 Multimodal posterior

We would like to assess the ability of FMPE to learn a multimodal posterior. For this purpose, we first set up a simple toy Gaussian mixture model.

Prior	$p(\theta) = \mathcal{N}(\theta \mu_0, \sigma_0^2 I)$
Simulator	$p(x \theta) = \sum_{i=1}^m w_i \mathcal{N}(x \theta + \mu_i, \sigma_i^2 I)$
Dimensionality	$\theta \in \mathbb{R}^d, x \in \mathbb{R}^d$
Hyperparameters	$\mu_i \in \mathbb{R}^d, \sigma_i^2 \in \mathbb{R}_{>0}; i = 0, \dots, m$ $w_i > 0; i = 1, \dots, m, \text{ with } \sum_{i=1}^m w_i = 1$

Using the normal-normal conjugacy, we can derive an analytical solution to the posterior,

$$\begin{aligned}
p(\theta|x) &= \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\theta | \tilde{\mu}_i, \tilde{\sigma}_i^2 I) \\
\tilde{\sigma}_i^2 &= (\sigma_0^{-2} + \sigma_i^{-2})^{-1} \\
\tilde{\mu}_i &= \tilde{\sigma}_i^2 (\sigma_0^{-2} \mu_0 + \sigma_i^{-2} (x - \mu_i)) \\
\tilde{w}_i &= \frac{v_i}{\sum_{j=1}^m v_j} \\
v_i &= w_i \sigma_i \tilde{\sigma}_i^{-1} \exp \left\{ -\frac{1}{2} \left[\tilde{\sigma}_i^{-2} \tilde{\mu}_i^T \tilde{\mu}_i + \sigma_0^{-2} \mu_0^T \mu_0 + \sigma_i^{-2} (x - \mu_i)^T (x - \mu_i) \right] \right\}
\end{aligned}$$

which is clearly multimodal. Setting the following hyperparameter values,

$$\begin{aligned}
m &= 2, \\
\mu &= \begin{bmatrix} 0 & -2.5 & 2.5 \\ \vdots & \vdots & \vdots \\ 0 & -2.5 & 2.5 \end{bmatrix} \in \mathbb{R}^{d \times (m+1)}, \\
\sigma &= [2, \ 1, \ 1] \in \mathbb{R}_{>0}^{m+1}, \\
w &= [0.5, \ 0.5] \in \mathbb{R}_+^m, \\
x &= [0 \dots 0]^T \in \mathbb{R}^d,
\end{aligned}$$

we obtain a bimodal posterior given by

$$\tilde{\mu} = \begin{bmatrix} -2 & 2 \\ \vdots & \vdots \\ -2 & 2 \end{bmatrix}, \tilde{\sigma} = \left[\sqrt{\frac{4}{5}}, \sqrt{\frac{4}{5}} \right], \tilde{w} = [0.5, \ 0.5].$$

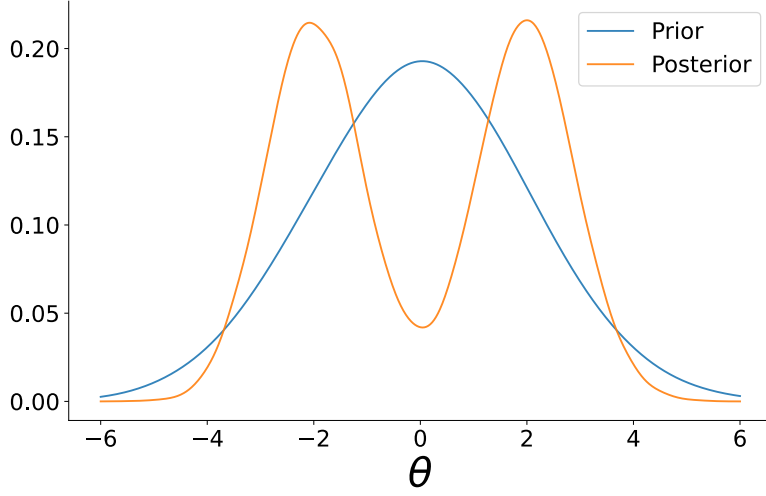


Figure 1: Histogram of prior $p(\theta)$ and posterior $p(\theta|x)$ densities for the specified hyperparameters. Each dimension $i = 1, \dots, d$ is identical and independent.

We can increase the number of dimensions d to increase the difficulty of the problem. While each dimension is identical and independent, the overall size of the space increases exponentially. For example, a simple random walk Metropolis-Hastings MCMC will already struggle at $d = 5$ with only ≈ 0.02 acceptance rate. We instead reference FMPE against an MCMC algorithm with a combination of global and local steps that explores the space more efficiently.

While we don't do this here, it is also worth noting that by selecting the vector field v_t to be of the form

$$v_t(\theta) = \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i q_t^i(\theta) \left[\frac{\theta}{t} - \frac{\sigma_t(\theta - t\tilde{\mu}_i)}{t(\sigma_t^2 + (t\tilde{\sigma}_i)^2)} \right],$$

where $q_t^i(\theta) := \mathcal{N}(\theta | t\tilde{\mu}_i, \sigma_t^2 + (t\tilde{\sigma}_i)^2)$

$$q_t(\theta) := \sum_{i=1}^m \tilde{w}_i q_t^i(\theta)$$

$$\sigma_t := 1 - (1 - \sigma_{\min})t,$$

we can force the solution to take the form of a Gaussian mixture $q_1(\theta) = \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\theta | \tilde{\mu}_i, \tilde{\sigma}_i^2)$. We can then choose to only learn the parameters $\{(\tilde{\mu}_i, \tilde{\sigma}_i^2, \tilde{w}_i), i = 1, \dots, m\}$. For our derivation of this result, see appendix A.

4.1.1 Case $d = 1$

For the one dimensional case, we set the following training hyperparameters:

We can see the FMPE learns the multimodal posterior even with just a small neural network with 8k hyperparameters. We quantify the difference to the true posterior using the C2ST metric from the simulation-based inference (SBI) benchmark [Lueckmann et al.,

Dataset	n	10000
FMPE	$\sigma_{\min}$	0.001
	α (distribution of t)	0.2
Neural Net	architecture	MLP
	hidden layers	3
	width	64
	non-linearity	SELU
	batch size	32
	epochs	500
Adam optimizer	learning rate	0.0001
	β	(0.9, 0.999)

Table 1: FMPE training hyperparameters for multimodal model with $d = 1$. The neural network has 8k parameters in total.

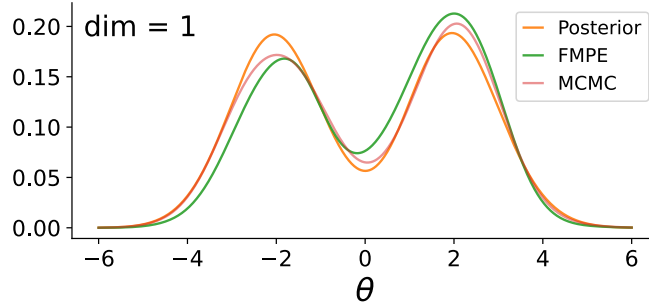


Figure 2: Histogram of posterior, trained FMPE approximation of the posterior, and a MCMC sample from the posterior.

2021]. This metric rates an approximation by the ability of a pre-specified NN classifier to distinguish between samples from the target and the approximation: $C2ST \in [0.5, 1]$, where lower is better. We report $C2ST_{\text{MCMC}} = 0.5080$ and $C2ST_{\text{FMPE}} = 0.5355$, confirming the the FMPE approximation is of essentially same quality as the MCMC sample.

4.1.2 Case $d = 5$

Keeping the dataset size fixed at $n = 10000$, we increase the number of dimensions to $d = 5$. For this problem, we set the following hyperparameters for training of the FMPE:

Dataset	n	10000
FMPE	$\sigma_{\min}$	0.001
	α (distribution of t)	0.2
Neural Net	architecture	MLP
	hidden layers	3
	width	256
	non-linearity	SELU
	batch size	32
	epochs	300
Adam optimizer	learning rate	0.0001
	β	(0.9, 0.999)

Table 2: FMPE training hyperparameters for multimodal model with $d = 5$. The neural network has 135k parameters in total.

At fixed $n = 10^4$, the FMPE approximation recognizes the bimodality of the distribution in each dimension, however, it fails to capture the sharpness of the two modes with only a shallow gap between. This is reflected in the C2ST metric, with $\text{C2ST}_{\text{MCMC}} = 0.521$ and $\text{C2ST}_{\text{FMPE}} = 0.6085$. If we increase the sample size to $n = 10^5$, however, we are once again able to fit the multimodal distribution perfectly. The achieved C2ST score then is 0.5470. The Figure showing this better fit can be found in Appendix B, Figure 6. The used neural network architecture remains the same.

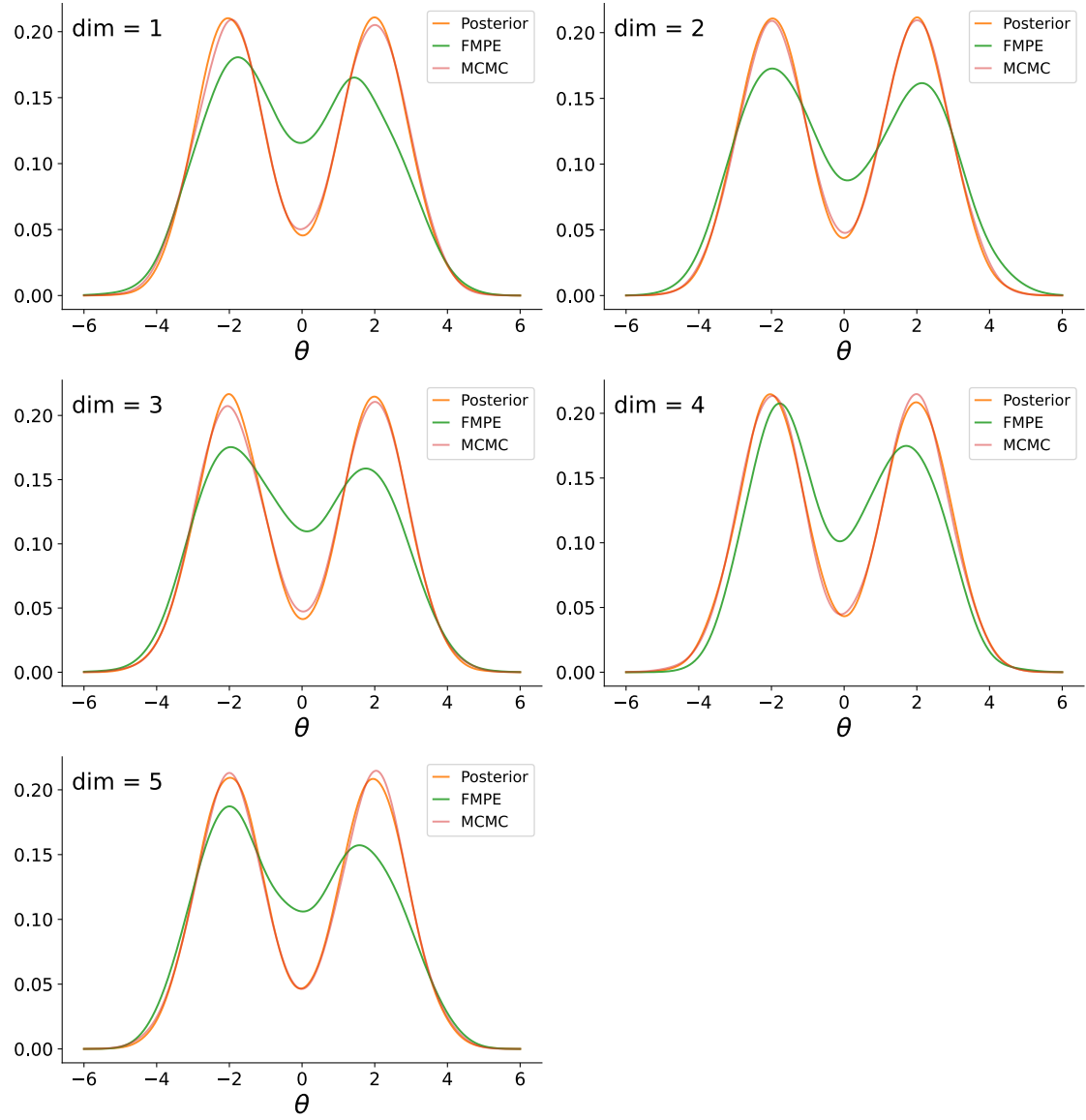


Figure 3: Histogram of posterior, trained FMPE approximation of the posterior, and a MCMC sample from the posterior for each one of the dimensions at $n = 10000$. FMPE recognizes multimodality but is easily discernible from the true posterior.

4.2 Radiocarbon dating

We further assess the ability of FMPE to learn a multimodal posterior on a non-symmetric real world example of radiocarbon dating. This example is originally from Nicholls and Jones [2001] but we use a simplified version from publicly available lecture notes (Section 1.4, Nicholls [2024]).

4.2.1 Observation model

A settlement in New Zealand has evidence of human activity, dated by radiocarbon dates from charcoal samples. The observed data are uncalibrated dates y_i with measurement uncertainty σ_i .

Sample id	i	y_i	σ_i
NZ 7758	1	580	47
NZ 7761	2	600	50
NZ 7757	3	537	44
NZ 7756	4	670	47
NZ 7755	5	646	47
WK 2589	6	630	35
NZ 7771	7	660	46

Table 3: Observed uncalibrated radiocarbon ages y_i of each charcoal sample and associated measurement uncertainty σ_i .

We calibrate the dates using the following observation model. We consider each uncalibrated age y_i as a noisy observation of unknown true age Θ_i ,

$$\begin{aligned} y_i &= \mu(\theta_i) + \epsilon_i \\ \epsilon_i &\sim \mathcal{N}(0, \sigma_c(\theta_i)^2 + \sigma_i^2) \end{aligned}$$

giving us a likelihood

$$p(y_i|\theta_i) = \mathcal{N}(y_i|\mu(\theta_i), \sigma_c(\theta_i)^2 + \sigma_i^2)$$

for each observation. The calibration map $\mu(\theta)$ and its uncertainty $\sigma_c(\theta)$ are obtained by radiocarbon dating labs [Hogg et al., 2013]⁶. We assume the observations to be independent, leading to an overall likelihood

$$p(y|\theta) = \prod_{i=1}^n p(y_i|\theta_i)$$

for $y = (y_1, \dots, y_n)$ and $\theta = (\theta_1, \dots, \theta_n)$.

⁶We follow the example exactly, using the calibration data provided [here](#).

4.2.2 Prior elicitation

Prior to observing the data, the archaeologists are confident the settlement must have been occupied within an interval $\Theta_i \in [L, U] = [500, 1000]$ years before the present (BP)⁷. Seeking to answer questions about the duration of the settlement, we aim to construct a prior non-informative with regards to that duration.

Let us consider the true settlement start date and end date ψ_2, ψ_1 respectively, such that $L < \psi_1 < \psi_2 < U$, and let the true specimen dates come from a Poisson process of constant rate λ over the interval $[\psi_1, \psi_2]$. Conditioning on the number of observations n , the true specimen ages θ_i are then uniformly distributed $\theta_i \sim \mathcal{U}[\psi_1, \psi_2]$, meaning

$$p(\theta|\psi) = \frac{1}{(\psi_2 - \psi_1)^n} \mathbb{I}(\psi_1 < \theta_1, \dots, \theta_n < \psi_2).$$

Enforcing uniform distribution of the span $\psi_2 - \psi_1$, we must have

$$p(\psi) \propto \frac{1}{(U - L - (\psi_2 - \psi_1))},$$

leading to an overall prior $p(\theta, \psi) = p(\psi)p(\theta|\psi)$

$$p(\theta, \psi) \propto \frac{1}{(\psi_2 - \psi_1)^n} \frac{1}{(U - L - (\psi_2 - \psi_1))}.$$

The posterior we shall simulate with MCMC and approximate with FMPE thus lies in an $n + 2$ dimensional space $\Omega = \{(\theta, \psi) \text{ in } [L, U]^{n+2} | \psi_1 < \theta_i < \psi_2, i = 1, \dots, n\}$.

4.2.3 Posterior estimation

For reference to the FMPE approximation, we employ a Metropolis Hastings MCMC that iteratively updates individual dimensions with a uniform proposal (in appropriate range) and a rescaling step [Nicholls, 2024].

As is visible on both Figures 4 and 5, the FMPE approximation generally learns the correct interval for each parameter but fails to learn the shape. No hint of understanding of the bimodal nature of the posterior is present. Additionally, the FMPE was not explicitly given any of the structure the prior has and thus assigns positive probability to events we disallowed. Specifically, the notion that no parameter can be less than $L = 500$ is not learned from the data. With the analytical posterior unavailable, we look at the C2ST metric directly between MCMC and FMPE. We report a C2ST score of 0.8030, which is in line with the visible difference between the two histograms in each of the dimensions.

⁷By convention, *the present* is taken to be the year 1950.

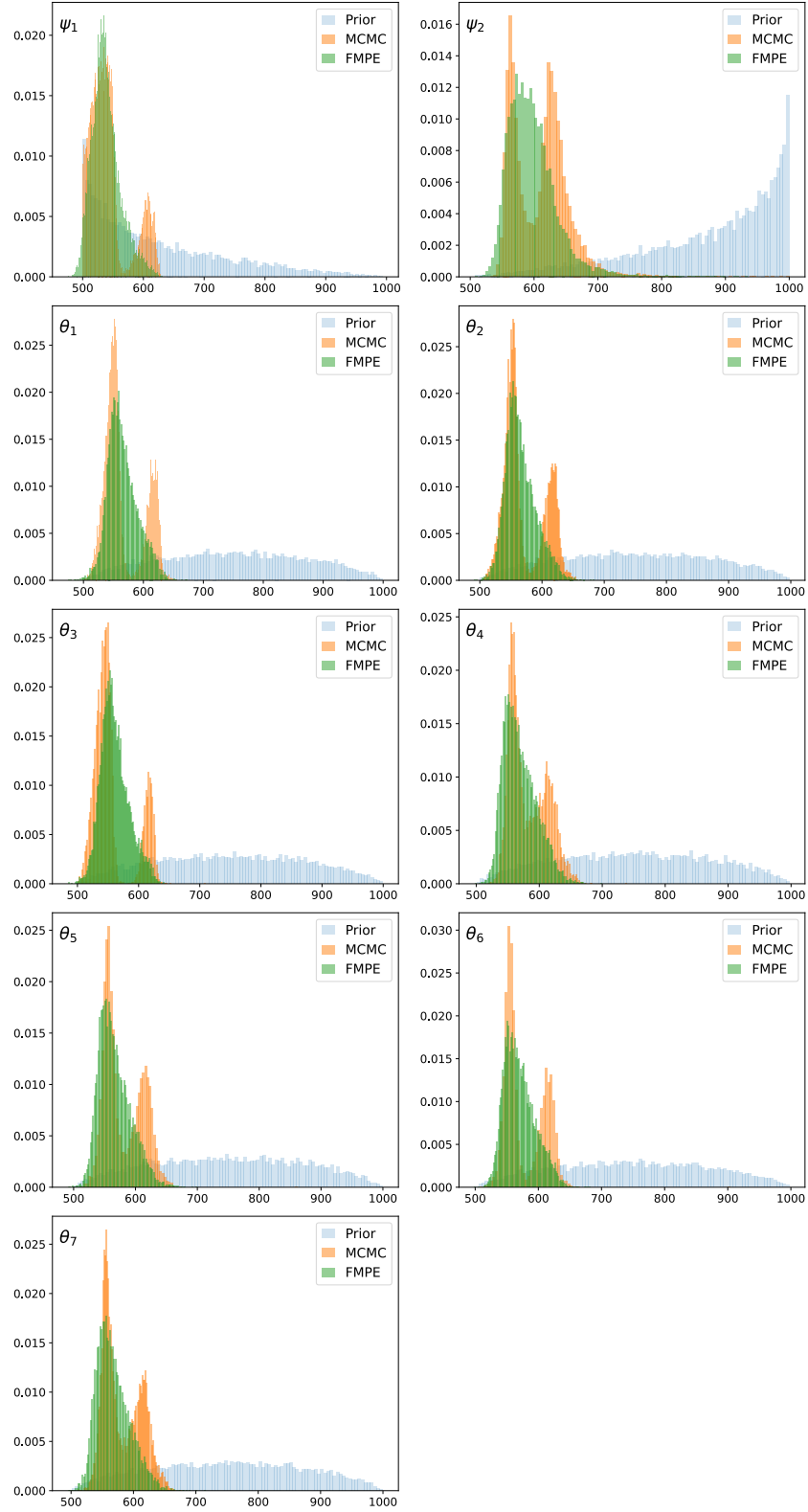


Figure 4: Histogram of posterior, trained FMPE approximation of the posterior, and a MCMC sample from the posterior for each one of the dimensions.

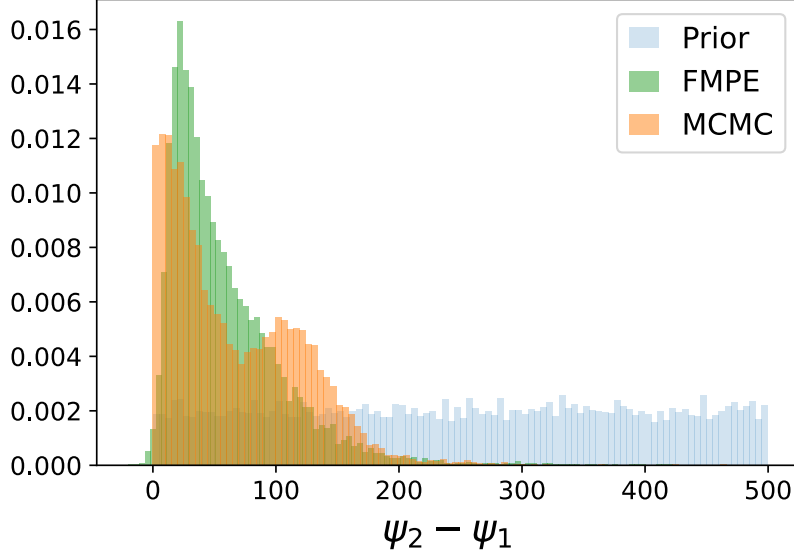


Figure 5: Histogram of prior, trained FMPE approximation of the posterior, and a MCMC sample from the posterior for the duration of the site’s settlement

Repeating the experiment, we have observed an improvement when scaling the dataset from $n = 10^4$ to $n = 10^5$. While the results at $n = 10^5$ (reported here) are still not satisfactory, we are fairly confident further increases in the model and dataset size would lead to yet better results. Due to personal computational constraints, we have not attempted these. It must also be noted that we have not attempted to use Tong et al. [2023] path designs, also due to computational limitations. These designs have shown promising results in the generative case, suggesting the possibility of improved performance for our problem as well.

Dataset	n	10^5
FMPE	$\sigma_{\min}$	0.0001
	α (distribution of t)	0.2
Neural Net	architecture	MLP
	hidden layers	3
	width	256
	non-linearity	SELU
	batch size	32
	epochs	300
Adam optimizer	learning rate	0.001
	β	(0.9, 0.999)

Table 4: FMPE training hyperparameters for radiocarbon dating. The neural network has 135k parameters in total.

5 Conclusion

In this thesis, we have explored the recent advancements of variational Bayesian inference. Specifically, we have investigated Flow Matching for Posterior Estimation and its ability to learn a multimodal posterior. We have seen promising results on a toy example of a mixture of Gaussians but poor performance on a more complex real problem of radiocarbon dating. Nevertheless, we remain optimistic about FMPE and its future applications. Our optimism is grounded in the ability of FMPE to scale well with added computational resources. Given the continuing presence of Moore’s law, the perception of what is computationally expensive will likely continue to evolve.

In the context of Bayesian inference, FMPE does not need be used as a standalone tool. With the ability to compute the learned approximate density, mass-covering properties (investigated by Wildberger et al. [2024]) and ease of sampling, the FMPE can very well serve as a high quality proposal distribution for MCMC or other stochastic algorithms. The FMPE, like any other amortizing method, initially requires significant computational cost to build an approximation of the posterior. However, once trained, it allows for much faster evaluations of the posterior across multiple sets of observations, making it highly efficient when dealing with repeated inference tasks. Additionally, the algorithmic pipeline is essentially identical and reusable across different inference problems with very little added effort for the researcher. To apply FMPE to a new problem, simply supply a sample from a new prior and new simulator and adjust the neural network to the new problem.

Beyond our experimental results, we have also provided a new proof for the optimum of Conditional Flow Matching, linking the objective to Decision Theory. Additionally, since the proof that FMPE learns to approximate the posterior has not been explicitly stated before, we give it here. Lastly, we extend a result of Wildberger et al. [2024] to show how FMPE can be forced into a solution of the form of a mixture of Gaussians. Beyond being just a curiosity, this result hints at the vastness of the CNF model and all the various distributions that can be learned within this framework.

In summary, Flow Matching for Posterior Estimation represents another advancement in the field of variational Bayesian inference, offering a powerful and flexible approach to approximating complex posteriors. While our research has demonstrate both the strength and current limitations of FMPE, more research is needed to assess the usefulness of FMPE in applied inference. Directions for future investigation include improving its performance on more complex real-world problems, exploring its potential as a proposal distribution for other simulation methods, and further investigating its theoretical properties. As computational resources continue to grow and evolve, the practical applications of FMPE are likely to expand, potentially changing how we approach Bayesian inference in various domains, from scientific modeling to machine learning.

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A Forcing FMPE to learn a mixture of Gaussians

Building upon the result in eq. 24, given by Wildberger et al. [2024], we here show how to force FMPE into learning a mixture of Gaussians

$$p(\theta) = \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\theta | \tilde{\mu}_i, \tilde{\sigma}_i^2)$$

with learnable parameters

$$\tilde{\mu}_i \in \mathbb{R}, \tilde{\sigma}_i^2 \in \mathbb{R}_{>0}, \tilde{w}_i > 0; \quad i = 1, \dots, m \text{ with } \sum_{i=1}^m \tilde{w}_i = 1.$$

The results here given for a one-dimensional Gaussian can be directly applied to the isotropic Gaussians used in Section 4.1. We employ the conditional optimal transport path with $\sigma_t(\eta) = 1 - (1 - \sigma_{\min})t$ and $\mu_t(\eta) = t\eta$. Additionally, let us use the following two convolution identities (denoted by $*$):

$$g_{\mu, \sigma^2}(\theta) := \theta \mathcal{N}(\theta | \mu, \sigma^2) \quad (25)$$

$$\mathcal{N}(\theta | \mu_a, \sigma_a^2) * \mathcal{N}(\theta | \mu_b, \sigma_b^2) = \mathcal{N}(\theta | \mu_a + \mu_b, \sigma_a^2 + \sigma_b^2) \quad (26)$$

$$g_{0, \sigma_a^2}(\theta) * \mathcal{N}(\theta | \mu_b, \sigma_b^2) = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_b^2} \left(g_{\mu_b, \sigma_a^2 + \sigma_b^2}(\theta) - \mu_b \mathcal{N}(\theta | \mu_b, \sigma_a^2 + \sigma_b^2) \right) \quad (27)$$

$$= \frac{\sigma_a^2(\theta - \mu_b)}{\sigma_a^2 + \sigma_b^2} \mathcal{N}(\theta | \mu_b, \sigma_a^2 + \sigma_b^2) \quad (28)$$

We find the marginal probability density to be

$$\begin{aligned} q_t(\theta) &= \int q_t(\theta | \eta) p(\eta) d\eta \\ &= \int \mathcal{N}(\theta | t\eta, \sigma_t^2) \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\eta | \tilde{\mu}_i, \tilde{\sigma}_i^2) d\eta \\ &= \sum_{i=1}^m \tilde{w}_i \int \mathcal{N}(\theta - t\eta | 0, \sigma_t^2) \mathcal{N}(t\eta | t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) t d\eta \\ &= \sum_{i=1}^m \tilde{w}_i \int \mathcal{N}(\theta - \eta' | 0, \sigma_t^2) \mathcal{N}(\eta' | t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) d\eta' \\ &= \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\theta | t\tilde{\mu}_i, \sigma_t^2 + (t\tilde{\sigma}_i)^2) \\ &= \sum_{i=1}^m \tilde{w}_i q_t^i(\theta) \\ \text{for } q_t^i(\theta) &= \mathcal{N}(\theta | t\tilde{\mu}_i, \sigma_t^2 + (t\tilde{\sigma}_i)^2), \end{aligned}$$

where we used the substitution $\eta' = t\eta$ and in the last step we applied the convolution equality given in 26. Now, we calculate the marginal vector field

$$\begin{aligned}
u_t(\theta) &= \int u_t(\theta|\eta) \frac{q_t(\theta|\eta)p(\eta)}{q_t(\theta)} d\eta \\
&= \frac{1}{q_t(\theta)} \int \frac{\eta - (1 - \sigma_{\min})\theta}{\sigma_t} \mathcal{N}(\theta|t\eta, \sigma_t^2) \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\eta|\tilde{\mu}_i, \tilde{\sigma}_i^2) d\eta \\
&= \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i \int \frac{t\eta - (1 - \sigma_{\min})t\theta}{t\sigma_t} \mathcal{N}(\theta - t\eta|0, \sigma_t^2) \mathcal{N}(t\eta|t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) t d\eta \\
&= \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i \int \frac{-(\theta - \eta') + (1 - (1 - \sigma_{\min})t)\theta}{t\sigma_t} \mathcal{N}(\theta - \eta'|0, \sigma_t^2) \mathcal{N}(\theta - (\theta - \eta')|t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) d\eta' \\
&= \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i \int \frac{-\eta'' + (1 - (1 - \sigma_{\min})t)\theta}{t\sigma_t} \mathcal{N}(\eta''|0, \sigma_t^2) \mathcal{N}(\theta - \eta''|t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) d\eta'' \\
&= \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i \left[\frac{-1}{t\sigma_t} g_{0, \sigma_t^2}(\theta) * \mathcal{N}(\theta|t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) + \frac{\theta}{t} \mathcal{N}(\theta|0, \sigma_t^2) * \mathcal{N}(\theta|t\tilde{\mu}_i, (t\tilde{\sigma}_i)^2) \right] \\
&= \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i \left[\frac{-1}{t\sigma_t} \frac{\sigma_t^2(\theta - t\tilde{\mu}_i)}{\sigma_t^2 + (t\tilde{\sigma}_i)^2} \mathcal{N}(\theta|t\tilde{\mu}_i, \sigma_t^2 + (t\tilde{\sigma}_i)^2) + \frac{\theta}{t} \mathcal{N}(\theta|t\tilde{\mu}_i, \sigma_t^2 + (t\tilde{\sigma}_i)^2) \right] \\
&= \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i q_t^i(\theta) \left[\frac{\theta}{t} - \frac{\sigma_t(\theta - t\tilde{\mu}_i)}{t(\sigma_t^2 + (t\tilde{\sigma}_i)^2)} \right],
\end{aligned}$$

where $\eta'' = \theta - \eta'$ and both convolution identities are used to get rid of the integral. By selecting the vector field v_t to be of the form

$$v_t(\theta) = \frac{1}{q_t(\theta)} \sum_{i=1}^m \tilde{w}_i q_t^i(\theta) \left[\frac{\theta}{t} - \frac{\sigma_t(\theta - t\tilde{\mu}_i)}{t(\sigma_t^2 + (t\tilde{\sigma}_i)^2)} \right], \quad (29)$$

we can thus force the solution to take the form $q_1(\theta) = \sum_{i=1}^m \tilde{w}_i \mathcal{N}(\theta|\tilde{\mu}_i, \tilde{\sigma}_i^2)$, where we learn the parameters $\{(\tilde{\mu}_i, \tilde{\sigma}_i^2, \tilde{w}_i), i = 1, \dots, m\}$. This result is of course exact only when taking the limit $\sigma_{\min} \downarrow 0$.

B Additional Figures

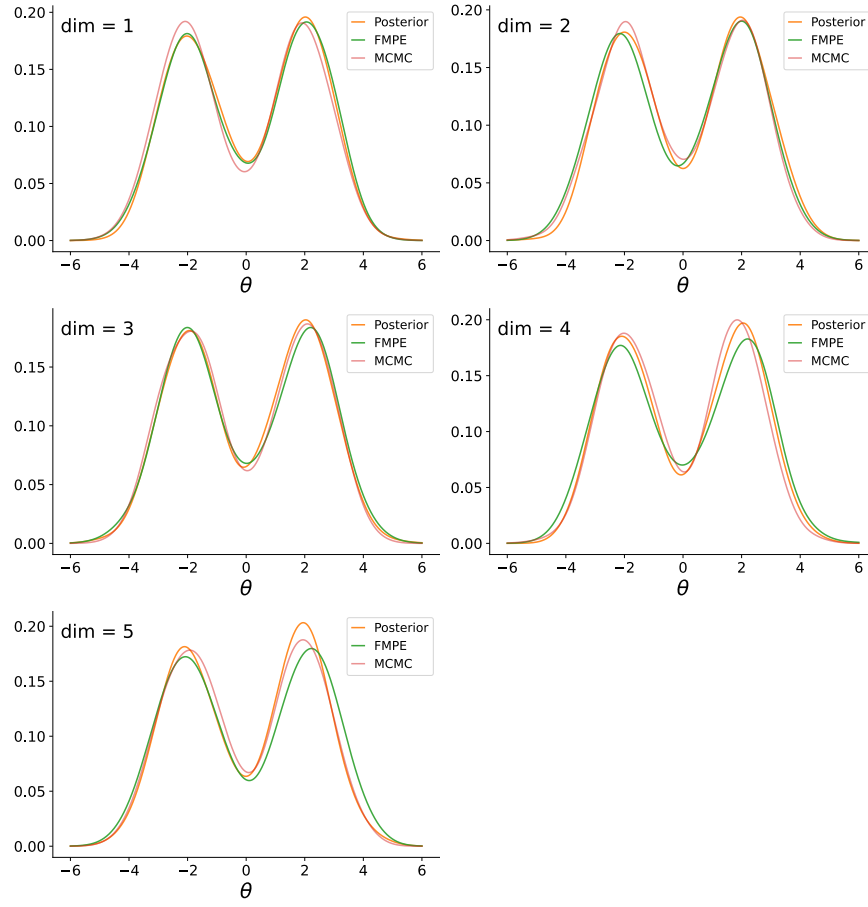


Figure 6: Histogram of posterior, trained FMPE approximation of the posterior, and a MCMC sample from the posterior for each one of the dimensions at $n = 10^5$. FMPE achieves great fit, with a C2ST score of 0.5470.

C Code

For reproducibility, our entire code implementation including the experiments can be found in the publicly available Github repository:

https://github.com/EmilJavurek/MSc_Thesis

For brevity, here we only show the main training loop and the Flow Matching function:

```
def FM_(x1, sigma_min, batch, t_dist):
    """
    Calculates the Flow Matching objective,
    for a batch of points from target

    Use to create:
    FM = partial(FM_, --here specify all inputs except x1--)
    And in actual training loop use FM()
    """
    t = t_dist(batch).view([batch] + [1] * (x1.dim() - 1))
    mu_t = t * x1
    sigma_t = 1 - (1-sigma_min)*t
    eps = torch.randn_like(x1)
    xt = mu_t + sigma_t * eps
    ut = (x1 - (1 - sigma_min) * xt) / (1 - (1 - sigma_min) * t)
    input = torch.cat([xt, t], dim = -1)
    return input, ut

#Training Loop
for epoch in tqdm(range(Epochs)):
    model.train()
    for theta_and_x in tr_dataloader:
        theta = theta_and_x[:, :theta_dim]
        x = theta_and_x[:, theta_dim:]

        #forward pass
        input, ut = FM(theta)                #input is [theta_t, t]
        full_input = torch.cat([x, input], dim = -1) #[x, theta_t, t]
        vt = model(full_input)
        loss = mse_loss(vt, ut)

        #backward pass
        optimizer.zero_grad()
        loss.backward()

        #gradient update
        optimizer.step()
```